PHYSIGYM: Bridging Agent-Based Biological Simulation and Reinforcement Learning, with an Observation-Space Study on Tumor Microenvironment Control

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Abstract

We present PHYSIGYM, a framework that exposes agent-based biological simulations as standard reinforcement learning (RL) environments. It connects the agent-based modelling framework PHYSICELL to the Gymnasium API, so that RL strategies can be tested on *in silico* biological processes. We illustrate it with a simplified tumor microenvironment (TME) toy model, in which a Soft Actor-Critic agent learns targeted drug-delivery policies that move the microenvironment toward an anti-tumoral state and, in the best episodes, nearly eradicate the tumor. We use the same case study to compare a family of observation spaces, from a partially observed macrophage-count baseline, through spatial scalar summaries, to multi-channel spatial images. With training on a structured rectangle layout and testing on a held-out network-field layout, image observations generalize markedly better than scalar summaries. Image modes reach a mean held-out test return of $\approx+32$ to $+33$, spatial-scalar summaries train comparably ($+43$) but fall to ≈ 0 out of distribution, and the macrophage-only POMDP baseline does not learn (-32). When the two layouts are swapped, the gap disappears: images protect the agent against overfitting a narrow training distribution. The observation space, which PHYSIGYM makes configurable, therefore determines policy quality and generalization in spatially structured environments. PHYSIGYM is open-source at `https://anonymous.4open.science/r/PhysiGym-76A7`.

1 Introduction

Biological systems are complex, dynamic, and often difficult to study in real life. In silico modeling aims to mimic these systems, providing a controlled environment for exploring specific aspects of biological processes and testing hypotheses. Agent-based models (ABMs) are well suited to biological systems in which agents (e.g. cells) interact according to predefined rules. They are used in oncology and immunology to study emergent behaviors, and they can suggest novel therapeutic strategies [Johnson et al., 2025, Laubenbacher et al., 2024, Rocha et al., 2024].

Ordinary differential equations (ODEs) and agent-based models (ABMs) are widely used to simulate the tumor microenvironment (TME). ODEs provide compact system-level descriptions, while ABMs capture spatial and cellular heterogeneity at higher computational cost. For dynamic treatment regimes (DTRs) [Gray and Pallavicini, 1982], however, simulation alone is insufficient: designing adaptive, sequential interventions requires a control framework. Reinforcement learning (RL), where an RL agent learns policies through interaction with an environment, offers such a framework [Barto et al., 1989]. By modeling treatment regimes as a sequence of decisions, RL enables adaptive control

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of biological simulations [Chakraborty and Murphy, 2014]. Efficient experimentation and policy learning also require a standardized interface. We therefore introduce PHYSIGYM, an interface that connects PHYSICELL [Ghaffarizadeh et al., 2018], an ABM framework for multicellular simulations of tissues or cell cultures (tumor microenvironment, other disease tissues, or bacteria colonies), with Gymnasium [Towers et al., 2024], a widely adopted RL application programming interface (API). PHYSICELL supports multiscale models of biological systems and couples them to the BioFVM diffusion transport solver [Ghaffarizadeh et al., 2016]. Gymnasium provides a common structure for training and evaluating RL policies.

The first implementation of RL on an ABM framework based on BioFVM was proposed by Zade et al. [2020], which applied Q-learning [Watkins and Dayan, 1992] to optimize Temozolomide treatment regimes for patients with glioblastoma multiforme. More recently, Aif et al. [2025] used PHYSICELL as a complex (high-fidelity) simulation environment for tumor growth, combined with a simple (low-fidelity) RL training environment of coupled ordinary differential equations (ODE) to learn treatment strategies for solid tumors.

Both studies couple RL to a simulator for a *single, fixed* problem. The coupling is bespoke, the observation and action spaces are tied to the task, and the code cannot be reused for another model. A practitioner who wants to ask a different question (a new cell type, a different readout, an image observation in place of a scalar one) must re-engineer the interface between simulator and RL each time. PHYSIGYM addresses this complementary need. It is a *general, reusable* bridge that turns *any* PHYSICELL model into a Gymnasium environment through subclassing and PHYSICELL's native user-parameter and custom-data constructs (Section 2.1). Observation space, action space, and reward become configuration choices that require no re-implementation. The observation-space study in this paper depends on this property: the same model is exposed in nine different ways with no modification of the simulator.

PHYSIGYM bridges PHYSICELL and Gymnasium to provide a reproducible and scalable framework for integrating agent-based simulations of biological systems with RL strategies. We present the motivation and design of PHYSIGYM, and show how it connects ABMs and RL to control complex biological systems. We focus on the tumor microenvironment, a biological system composed of cancer cells and surrounding immune and stromal cells. Their interactions are known to determine the response to immunotherapy, a therapeutic approach used in an increasing number of cancer types.

To illustrate the framework, we build a simplified TME toy model and use it as an end-to-end RL case study. The model also serves as a controlled testbed for a question that arises in any spatially explicit biological environment: among the observation spaces PHYSIGYM can expose, which ones help an RL agent learn an effective dynamic treatment regime? When the agent trains on a narrow, structured layout, image observations that preserve the spatial layout of cells and substrates generalize to held-out layouts, and compact scalar summaries do not. When it trains on irregular layouts, the two perform alike. This finding concerns one deliberately simplified model. The primary contribution of this work is PHYSIGYM.

Contributions. (1) PHYSIGYM: an open-source framework that bridges the PHYSICELL ABM and the Gymnasium RL API, turning any PHYSICELL model into an RL environment with minimal code changes. (2) A simplified TME toy model demonstrating an end-to-end RL case study in which an agent learns a spatial drug-delivery policy that drives the microenvironment toward an anti-tumoral state. (3) A systematic comparison of observation spaces on the toy model, in both transfer directions, showing that spatially explicit (image) representations protect the agent against overfitting a narrow training distribution, and that this advantage disappears when the training distribution is irregular.

2 Methods

2.1 PHYSIGYM design and implementation

PHYSICELL is an open-source ABM framework written in C++ that models multicellular systems using classical mechanics and the BioFVM diffusive transport solver [Ghaffarizadeh et al., 2018, 2016]. Cells are agents governed by type-specific rule sets; substrates such as oxygen or cytokines diffuse through the domain. Intracellular signalling networks [Letort et al., 2018, Ponce-de Leon et al., 2023, Ruscone et al., 2024] and extracellular matrix fibres [Noël et al., 2024] can be added, yielding ABMs that are spatially explicit in 2D or 3D, off-lattice, and multiscale [Metzcar et al., 2019].

RL frames sequential decision-making as a Markov Decision Process (MDP) [Bellman, 1957, Puterman, 2014] with state space S, action space A, transition model T, and reward R. The objective is

$$\arg\max_{\pi \in \Pi} \mathbb{E}\left[\sum_{t=0}^{\tau} \gamma^t r_t \mid s_0,\, \pi\right].$$

In our setting the agent never sees the full simulator state (every cell's position and internal variables, every voxel's substrate concentrations); it receives an observation $o_t = f(s_t)$ from an observation space O chosen by the user, so the control problem is formally a partially observed MDP (POMDP). The observation map f is the component we vary in Section 2.4. Gymnasium [Towers et al., 2024] is the standard Python interface for MDPs, and it promotes modularity and code reuse across algorithms and environments.

PHYSIGYM connects these two ecosystems by extending the Python interpreter [van Rossum, 1995, Python Software Foundation, 2024] with PHYSICELL. The C++ simulation loop is split into three Python-callable functions: `physicell_start`, `physicell_step`, and `physicell_stop`. For most models only `physicell_step` needs to be modified (e.g. to inject a drug). The read functions `get_cell` and `get_microenv`, implemented in `physicellmodule.cpp`, return cell positions and substrate concentration fields. The template models of PHYSIGYM also provide `get_graph`, which returns cell neighbourhood graphs. The build used for the experiments below does not expose it, and none of our observation modes needs it.

The interface between PHYSICELL and Gymnasium uses PHYSICELL's built-in *user parameter* and *custom data* constructs, configurable through PhysiCell Studio [Heiland et al., 2024] without recompilation. Users subclass `CorePhysiCellEnv` in `physicell_model.py` to specify observation space, action space, and reward; the base class handles all model-independent Gymnasium bookkeeping. Any existing PHYSICELL model can therefore become a PHYSIGYM environment with minimal code changes. PHYSIGYM is tested on Linux, WSL, and macOS, with continuous integration via GitHub Actions. Documentation and tutorials are at `https://anonymous.4open.science/r/PhysiGym-76A7/man`.

Fidelity and reproducibility. We compared one run through PHYSIGYM with the same run in standalone PHYSICELL (same seed and initial condition, one thread, no drug), matching frames at identical simulated times (Appendix B). Cell counts agree exactly for the first 495 simulated minutes. Positions differ by 7.1×10^{-15} μm after the first step, and this difference grows by approximately one order of magnitude every 45 minutes, the signature of a chaotic system amplifying a rounding difference. Two standalone runs, by contrast, are bit-identical over $12{,}002$ frames. Across separate processes, single-threaded PHYSIGYM runs reproduce bit for bit, while runs with 8 OpenMP threads diverge within 60 steps. Our results are therefore reproducible as distributions over seeds (curves, means, and tests), and individual trajectories reproduce only in single-threaded execution.

Figure 1 shows the resulting dataflow. Listing 2 gives a complete minimal environment, in which the user overrides only the observation, action, and reward, and the base class handles the PHYSICELL start/step/stop loop.

2.2 Implementation of a Tumor Microenvironment Toy Model

We developed a simplified tumor microenvironment (TME) *toy model* to illustrate the potential uses of PHYSIGYM. The model is deliberately small and computationally cheap, so that RL agents can be trained to convergence many times (across observation spaces and seeds). It keeps the spatial, multicellular, and chemically coupled structure that distinguishes agent-based models (ABMs) from ODE surrogates. It includes three cell types, five diffusible substrates, and a set of simple agent-based rules. The agent controls tumor progression by modulating the whole microenvironment, and it has no means of killing cancer cells directly.

Domain and Dynamics. The continuous-space environment is simulated on a $64 \times 64\,\mu$m 2D domain. The RL gym step interval is $\Delta t = 15$ min, with episodes running up to $10{,}080$ min (7 simulated days).

Cell populations and control problem. The control problem is highly non-linear and indirect. The agent's action (deploying a localized therapeutic, `drug_1`) has zero direct cytotoxicity. The agent must learn to modify the transition dynamics of the environment by reversing a spatial immunosuppression loop. Three interacting cell populations move along biochemical gradients (chemotaxis):

- **Tumor cells (target):** immobile cells that divide continuously. They secrete a chemical signal (`tumor_molecule`) that suppresses the local immune response.
- **Macrophages (mediators):** mobile, immortal cells that migrate toward the tumor. Their behavior depends on their state: the tumor's chemical signal converts them from an anti-tumoral to a pro-tumoral state.
- **T-cells (effectors):** mobile, immortal cells that can trigger tumor apoptosis. Pro-tumoral macrophages strongly repel them.

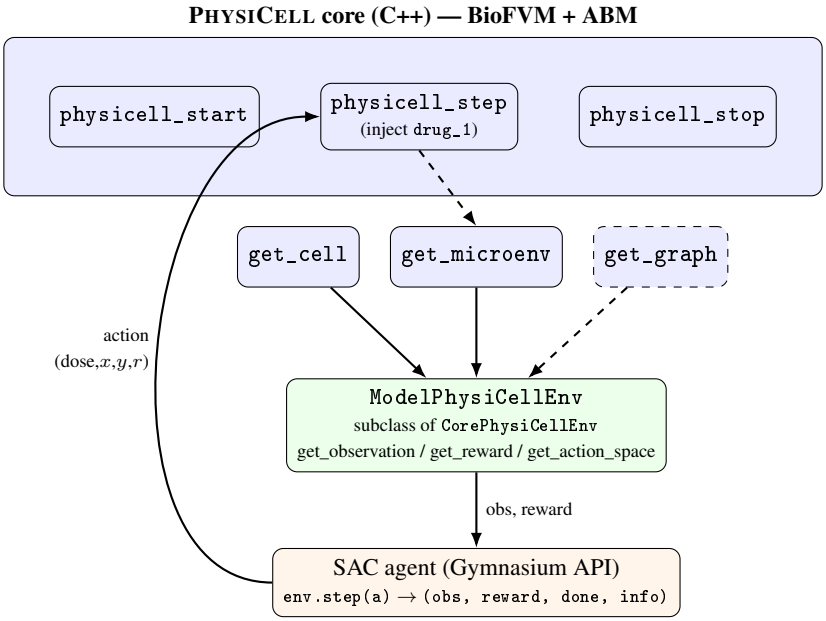


Figure 1: PHYSIGYM dataflow. The PHYSICELL C++ loop is exposed to Python as three callables (`physicell_start/step/stop`); only `physicell_step` is modified per model (here, to inject `drug_1`). Read functions surface cell positions (`get_cell`) and substrate fields (`get_microenv`). `get_graph` (dashed) returns neighbourhood graphs in the template models and is not used here. A user subclasses `CorePhysiCellEnv` to assemble these into a Gymnasium observation/reward, and any RL algorithm drives the environment through the standard `step`/`reset` API.

The mechanism is a simplified abstraction of *macrophage-repolarising immunotherapy*. Tumour-associated macrophages (TAMs) with a pro-tumoral, M2-like phenotype are abundant in many solid tumours and are associated with poor prognosis [Mantovani et al., 2017]; they can physically exclude $CD8^+$ T-cells from tumour nests and thereby limit checkpoint-blockade efficacy [Peranzoni et al., 2018]. Re-educating TAMs toward an anti-tumoral, M1-like state, for example with CSF-1R inhibition [Pyonteck et al., 2013] or TLR7/8 agonists delivered by nanoparticles that accumulate in TAMs [Rodell et al., 2018], is an active therapeutic strategy. `drug_1` plays the role of such a TAM-repolarising agent: a locally delivered immunomodulator with no direct cytotoxicity. We use this mechanism qualitatively; the rates in the toy model are not fit to any of these studies (see Limitations).

The result is a self-reinforcing feedback loop: the tumor protects itself by converting macrophages, which then block T-cells from reaching it. As illustrated in Figures 3 and 4, the agent maximizes reward by learning where to deploy the drug so that it re-programs macrophages to an anti-tumoral state. This intervention modifies the chemical gradients and opens a path for T-cells to reach the tumor and kill it. `drug_1` does not diffuse: it acts only on the disc where the agent injects it, and the largest disc the action space permits covers 6.3% of the domain. The repolarised macrophages emit a diffusing anti-tumoral factor, which forms the gradient the T-cells follow. Where the drug is placed therefore decides which macrophages are repolarised, and the agent must aim.

2.3 Implementation of an RL approach to control the TME

Our objective is complete tumor eradication with minimal drug exposure. We describe the action space A, the reward R, and the training and test distributions. The observation space S is the component we vary across experiments. Section 3 compares a family of observation modes, from a partially observed scalar baseline through spatial scalar summaries to multi-channel spatial images, to assess whether spatially explicit representations help the agent learn and generalize on this toy model.

```python
from physigym.envs import CorePhysiCellEnv
from gymnasium import spaces
import numpy as np

class ModelPhysiCellEnv(CorePhysiCellEnv):
    # Action: localized drug injection (dose, x, y, radius), all normalized.
    def get_action_space(self):
        box = lambda: spaces.Box(0.0, 1.0, (1,), np.float32)
        return spaces.Dict({"drug_1_dose": box(), "drug_1_x": box(),
                            "drug_1_y": box(), "drug_1_radius": box()})

    # Observation: e.g. per-cell-type 64x64 image channels (img_mc_cells).
    def get_observation_space(self):
        return spaces.Box(0, 255, (self.n_channels, 64, 64), np.uint8)

    def get_observation(self):
        # get_cell / get_microenv surface ABM state from the C++ core.
        return self.render_cell_channels() # -> (C, 64, 64) uint8

    # Reward: tumor suppression minus a drug-cost penalty (Eq. in paper).
    def get_reward(self):
        dc = (self.count_prev - self.count) / self.growth_norm
        return self.w_cell * dc - self.w_dose * self.get_dose_spent()

    def get_terminated(self):
        return self.count == 0 # tumor eradicated
```

Figure 2: A complete PHYSIGYM environment: subclass `CorePhysiCellEnv` and override the observation, action, and reward. The base class handles all model-independent Gymnasium bookkeeping (`reset`/`step`, the PHYSICELL start/step/stop loop, and the `get_cell`/`get_microenv` read-back), so no C++ recompilation is needed to expose a new state space. (Condensed from `physicell_model.py`.)

We train every configuration with Soft Actor-Critic (SAC) [Haarnoja et al., 2018], using an asynchronous parallel implementation (Appendix C); the network architectures and hyperparameters, including how they were selected, are given in Appendix C.3.

Evaluation protocol. Each run uses 13 parallel environments. Every fourth episode of each environment is a *test* episode: its initial condition is freshly sampled from the held-out network-field distribution, and the agent's actions do not enter the replay buffer. All other episodes are training episodes on freshly sampled rectangle layouts. A run therefore collects on the order of 200 held-out test episodes, each on a distinct initial condition, and the per-run test return we report is the exponentially weighted mean (span 50) over the last test episodes at the end of training. Each observation mode is trained with $n = 5$ independent training seeds, and all uncertainty statements aggregate over these seeds (bootstrap CIs, Appendix D; significance tests, Appendix E).

Action space. At each gym step the agent selects $a_t = (\text{dose}, x, y, r) \in [0, 1]^4$, where dose is the drug concentration applied, (x, y) is the injection centre (normalised domain coordinates), and r is the injection radius (normalised by the domain diagonal).

Reward. The reward is a normalised tumor-suppression signal minus a weighted drug-cost penalty:

$$r_t \;=\; w_{\text{cell}}\,\frac{c_{t-1} - c_t}{c_{t-1}(e^{\lambda \Delta t} - 1)} \;-\; w_{\text{dose}}\, d_t, \tag{1}$$

where c_t is the alive tumor-cell count at step t, λ is the intrinsic growth rate (from `PhysiCell_settings.xml`), d_t is the amount of drug injected at step t divided by the domain volume, $w_{\text{cell}} = 0.3$, and $w_{\text{dose}} = 2$. Because the drug fills a disc, d_t equals the commanded dose times the fraction of the domain the disc covers. The denominator $c_{t-1}(e^{\lambda \Delta t} - 1)$ is the number of new tumor cells expected over one step from unchecked intrinsic exponential growth, so the ratio measures the observed change *relative to* the growth the agent had to overcome. This makes the tumor term scale-free in the absolute cell count and comparable across initial conditions with different tumor sizes: a value of $+1$ means one step's worth of expected growth was fully cancelled, independent of how many cells were present. Only the ratio $w_{\text{dose}}/w_{\text{cell}} \approx 6.7$ affects which policy is optimal. It

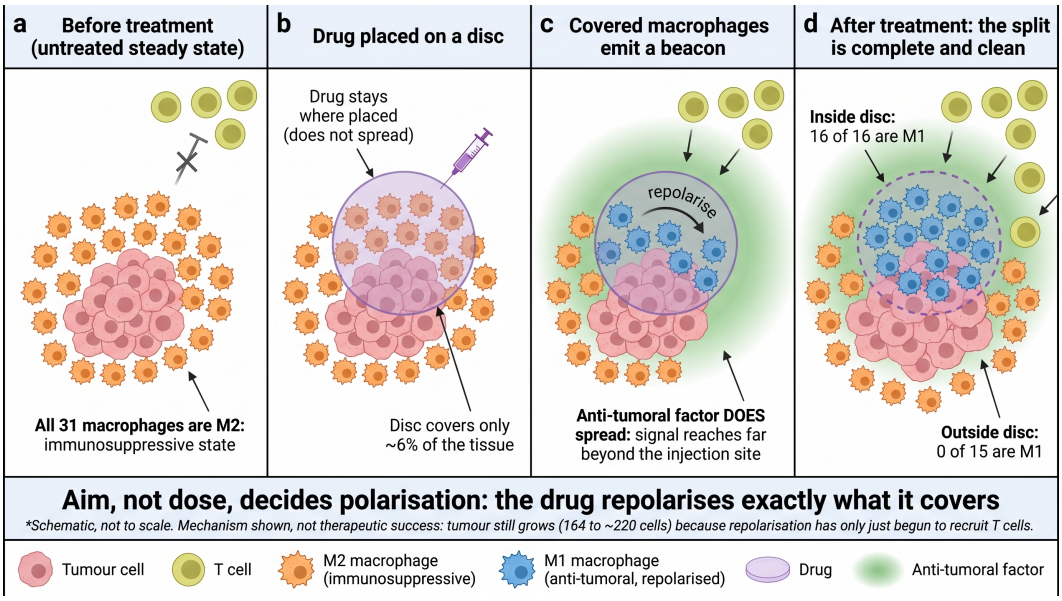


Figure 3: The indirect drug mechanism (schematic, not to scale). **(a)** Untreated steady state: all 31 macrophages are M2 and the T-cells stay outside the tumor. **(b)** The drug is injected on a disc and does not spread. **(c)** The covered macrophages repolarise to M1 and emit the anti-tumoral factor, which diffuses and attracts T-cells. **(d)** After treatment, 16 of 16 macrophages inside the disc are M1 and 0 of 15 outside it. Counts are from one simulation of the model. Created with BioRender.com.

makes a unit of dose expensive against a unit of normalised tumor control, so blanket dosing has to justify its cost.

Training and Test Distributions

We train and test on two *structurally distinct* spatial distributions, so that test performance measures out-of-distribution (OOD) generalisation to an unseen geometry:

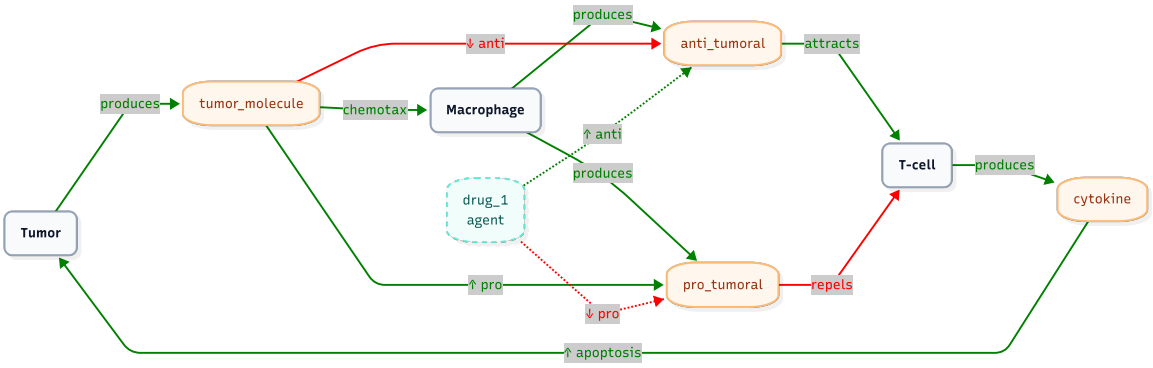


Figure 4: Tumor Microenvironment (TME) signalling network. Green solid arrows denote production, activation, or attraction, while red solid arrows indicate suppression or repulsion. The dashed pathways represent the intervention of the agent-controlled drug (`drug_1`), highlighting its role as an immune-modulator that repolarizes macrophages to indirectly drive T-cell mediated tumor apoptosis.

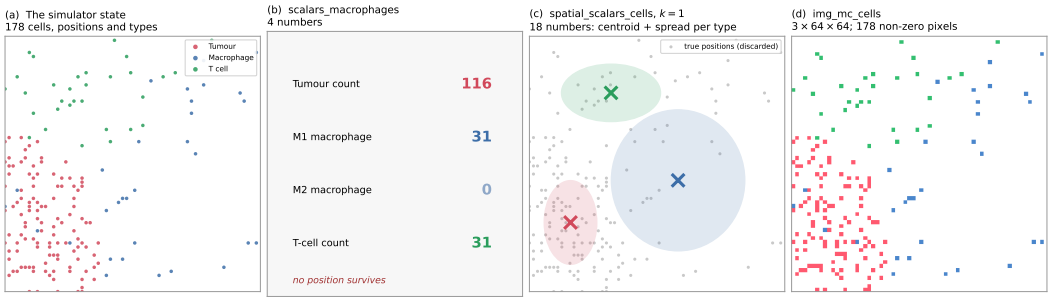


Figure 5: One held-out network-field initial condition seen by three observation modes. **(a)** Simulator state: every cell's position and type. **(b)** `scalars_macrophages`: four counts, with no position. At $t = 0$ both polarising fields are still empty, so the M1/M2 rule labels every macrophage M1. **(c)** Spatial-scalar summary with $k = 1$: a centroid ($\times$) and a spread per type. The positions in grey are discarded. The macrophages form several clumps, and their centroid falls in empty space. **(d)** `img_mc_cells`: three 64×64 occupancy channels, which keep every position.

- **Rectangle (training)**: Each cell type is placed uniformly in a narrow vertical strip, with non-overlapping strip positions (Appendix A). This is a highly structured, low-entropy layout.
- **Network-field (held-out test)**: Initial positions are sampled from a spatially correlated random field (Appendix A), producing irregular, clustered configurations never seen during training.

The transfer from the simple, regular rectangle to the richer, irregular network-field is demanding. A policy that exploits the fixed strip geometry of the rectangle cannot solve the network-field, so the test return measures whether an observation space encodes *transferable* structure. Figure 9 shows representative examples of both distributions. To quantify the shift, we describe each layout by 18 spatial statistics (per-population centroid, per-axis spread and anisotropy, and the three inter-population centroid distances) and compute the sliced Wasserstein-2 distance between 150 layouts of each family. The two families are at a distance of 1.27, against 0.19 ± 0.01 between two halves of the same family split by seed.

2.4 Observation space O: modes compared

The simulator can expose several kinds of data: some carry explicit spatial information, others are spaceless summaries (e.g. a small vector of global statistics). Because PHYSIGYM can build all of these from the same simulation, we use the toy model to study how the choice of observation space affects the dynamic treatment regime that RL learns. We compare nine modes, from a partially observed scalar baseline to multi-channel spatial images, on cumulative return and on generalization to unseen spatial configurations. Figure 5 shows one simulator state as seen by three of them.

Let $N_c = 3$ (tumor, T-cell, macrophage) and $N_s = 5$ substrates. Several modes optionally split the macrophage population into its *M1* (anti-tumoral) and *M2* (pro-tumoral) polarisation states, which we denote with the suffix `m1m2`; the classification rule is given below.

Macrophage polarisation (M1/M2). Macrophages do not carry an explicit polarisation label in the simulator; their state is implicit in the local chemical environment. We recover it from the microenvironment: for each alive macrophage we look up the two competing substrate fields at its nearest voxel and classify it as *M2* (pro-tumoral) if $\mathrm{conc}_{\texttt{pro_tumoral}} > \mathrm{conc}_{\texttt{anti_tumoral}}$ at that voxel, and *M1* (anti-tumoral) otherwise. At the first observation of an episode both fields are still zero, the strict inequality fails, and every macrophage is labelled M1 (Figure 5b). From the next step on, the label follows the fields. This one-frame artefact is present in all runs reported here. The pro-/anti-tumoral balance, which is the quantity the agent must reverse, thus becomes observable as a count (`scalars` modes) or as two spatial channels (`img` modes). Without this step it would remain latent in the substrate fields.

Scalar Observation Spaces

`scalars_macrophages` **(POMDP baseline):** Alive count per cell type, with the macrophage slot split into M1 and M2 counts; each is normalised to the initial population ($s = c/c_{\text{init}} - 1$). This mode exposes only *how many* macrophages are polarised each way. It contains no spatial information: neither cell positions nor substrate maps. It is therefore a strongly partially observed (POMDP) baseline. The state variables that govern the transition dynamics (positions of cells and gradients) are hidden, and the agent acts on a heavily compressed summary.
Space: $\mathbb{R}^{N_c+1}$

`spatial_scalars_cells_substrates`**:** Per-type normalised cell counts and per-substrate maxima, augmented with 6 k-means spatial statistics per cell type (cluster occupancy and centroids), giving a compact spatial summary without full images.
Space: $\mathbb{R}^{N_c+N_s+6kN_c}$

`spatial_scalars_cells_m1m2` / `..._substrates_m1m2`**:** As above but with the macrophage population split into M1/M2 for both the scalar counts and the k-means spatial features (`..._substrates_m1m2` additionally keeps the substrate maxima).
Space: $\mathbb{R}^{(N_c+1)+[N_s]+6k(N_c+1)}$

`spatial_scalars_cells_spatial_no_scalars_substrates_m1m2`**:** M1/M2 cell counts plus k-means spatial features for *both* cell types and substrate fields (spatially resolved substrate summaries in place of scalar maxima).
Space: $\mathbb{R}^{(N_c+1)+6kN_s+6k(N_c+1)}$

Image-Based Observation Spaces

`img_mc_cells` **(I1):** One 64×64 channel per cell type.
Space: $\mathbb{R}^{N_c\times64\times64}$ (`uint8`)

`img_mc_cells_m1m2`**:** I1 with the macrophage channel split into separate M1 and M2 spatial maps.
Space: $\mathbb{R}^{(N_c+1)\times64\times64}$ (`uint8`)

`img_mc_cells_substrates` **(I2):** I1 concatenated with all five substrate concentration maps, providing direct spatial feedback on drug distribution and the polarising factors.
Space: $\mathbb{R}^{(N_c+N_s)\times64\times64}$ (`uint8`)

`img_mc_cells_substrates_m1m2` **(full information):** I2 with the additional M1/M2 macrophage channels. This mode carries every signal the others expose (cell positions, substrate fields, and the explicit polarisation split) and serves as the near-fully-observed reference for the compressed modes.
Space: $\mathbb{R}^{(N_c+N_s+2)\times64\times64}$ (`uint8`)

An observability spectrum

Together these modes form an observability spectrum, from `scalars_macrophages`, a POMDP that sees only polarisation counts, to `img_mc_cells_substrates_m1m2`, which spatially resolves every state variable relevant to the control problem. The indirect drug mechanism (Section 2.2) requires multi-step credit assignment: the agent injects drug, macrophages repolarise over several steps, anti-tumoral gradients shift, T-cells migrate, and only then do tumor cells die. Scalar summaries cannot tell the agent whether an injection reached macrophages, whether polarisation occurred, or where T-cells will concentrate next. The per-pixel substrate and M1/M2 channels of the image modes show the polarisation state and the drug distribution in space at each step.

3 Results

3.1 Aggregated performance

Table 1 summarises final performance (seeds averaged). Figures 6 and 7 show learning curves vs. cumulative environment steps with mean ± std shading across seeds.

Table 1: Performance at end of training (train = rectangle, test = network-field, seeds averaged). Train$_\mu$/Test$_\mu$: mean EWMA-50 return, with a 95% bootstrap confidence interval on the mean ($B = 10{,}000$ seed resamples); n = number of seeds. This is the curated main set; the remaining `m1m2` scalar variants appear in Appendix D. RAND is a random-action policy (5 seeds), included as a learning-free reference: modes below it do no better than random behaviour.

ID	Mode	n	Train$_\mu$ (95% CI)	Test$_\mu$ (95% CI)
I1	`img_mc_cells`	5	$+30.8$ $[+21.3, +38.0]$	$+\mathbf{33.4}$ $[+23.6, +43.2]$
I2	`img_mc_cells_substrates`	5	$+45.3$ $[+40.5, +49.4]$	$+32.7$ $[+28.6, +36.7]$
I2m	`img_mc_cells_substrates_m1m2`	5	$+40.8$ $[+33.0, +47.3]$	$+32.5$ $[+24.4, +40.5]$
S3s	`spatial_scalars_cells_substrates`	5	$+42.9$ $[+24.7, +58.8]$	-0.0 $[-6.2, +9.1]$
RAND	random policy (baseline)	5	-18.4 $[-21.9, -14.9]$	-25.3 $[-28.2, -22.8]$
POMDP	`scalars_macrophages`	5	-20.4 $[-28.6, -13.8]$	-31.8 $[-33.4, -29.4]$

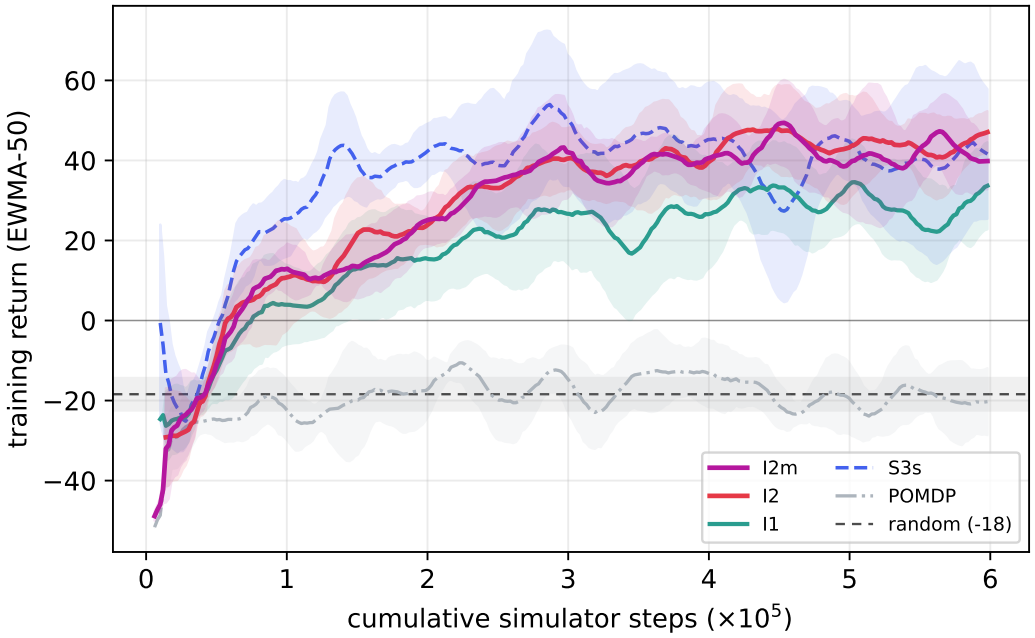


Figure 6: Training return on the rectangle layout (EWMA-50, mean $\pm 1\sigma$ across seeds) vs. cumulative *simulator* steps ($\times 10^5$; each of the 10^5 agent decisions is held for 6 simulator steps, giving a 6×10^5 budget). All modes learn on the training layout except the `scalars_macrophages` POMDP baseline, which flatlines near -20. Spatial-scalar summaries (S3s) train as fast as, or faster than, the image modes here. The separation appears only out of distribution (Figure 7).

3.2 Key findings

Separation under distribution shift. On the training rectangle layout the spatial-scalar summary S3s is fully competitive: its training return ($+42.9$) is on par with the best image mode I2 ($+45.3$). The modes separate only out of distribution. On the held-out network-field, the three image modes retain $\approx +32$ to $+33$ test return, while S3s falls to -0.0. Ranked on training fit alone, the scalar summary would appear among the best modes. The OOD test shows that it has overfit the fixed strip

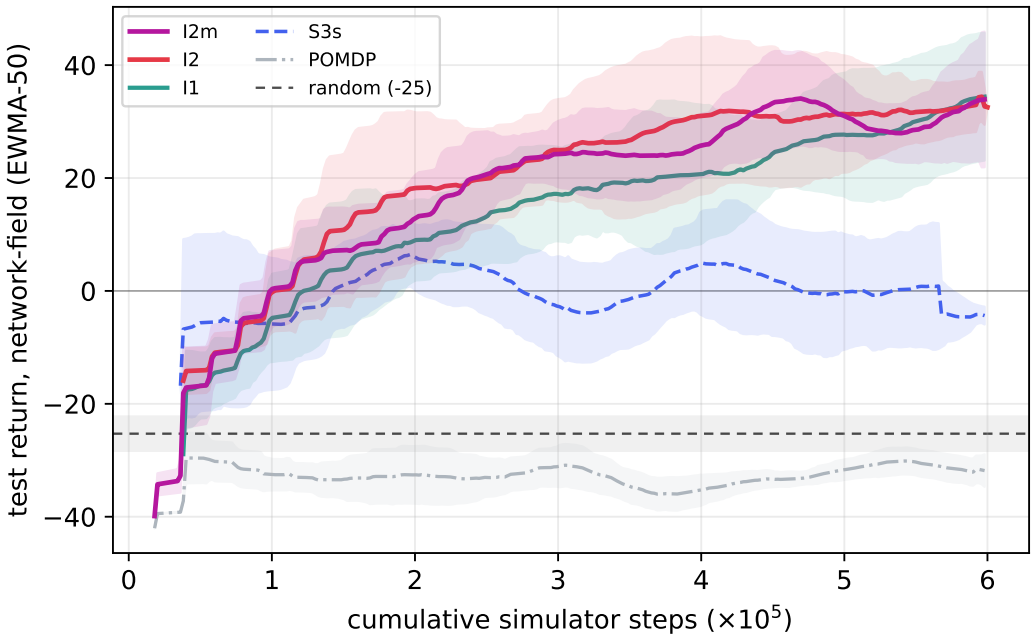


Figure 7: Held-out test return on the network-field layout (mean $\pm 1\sigma$ across seeds, EWMA-50) vs. cumulative simulator steps. The image modes (I2m, I2, I1) climb steadily to $\approx +33$. The spatial-scalar summary (S3s), which trained as well as the image modes (Figure 6), falls to ≈ 0: it has overfit the fixed rectangle geometry. The dashed line marks the random-policy baseline (test ≈ -25, shaded $\pm 1\sigma$). The `scalars_macrophages` POMDP baseline stays at or below it throughout, i.e. never beats random.

geometry of the rectangle. This separation is statistically supported: the image modes' 95% bootstrap confidence intervals on the mean test return lie entirely above $+23$ and do not overlap those of any spatial-scalar mode, which all reach down to $+0.4$ or below (Table 1; all nine modes in Table 5). Pooling the per-seed test returns of the three image modes against the four spatial-scalar modes, the image family is higher with a Mann–Whitney U-test $p = 6 \times 10^{-7}$ and a Cliff's $\delta = +0.97$ (near-complete rank separation), so the gap is not an artefact of seed selection (Appendix E).

Transfer of image observations. All three image modes, `img_mc_cells` (I1, $+33.4$), `img_mc_cells_substrates` (I2, $+32.7$), and the full-information `img_mc_cells_substrates_m1m2` (I2m, $+32.5$), generalize to the network-field, and their confidence intervals overlap. Since I1 already generalizes well, preserving the *spatial layout* of cells is the primary driver of transfer in this toy model. The extra substrate and M1/M2 channels of I2/I2m do not further separate on average.

POMDP baseline and random policy. `scalars_macrophages`, which observes only M1/M2 counts and no spatial information, does not improve on either distribution (train -20.4, test -31.8) and does *not* beat a random-action policy (train -18.4, test -25.3; RAND in Table 1). Its test return is, if anything, below random, and the two bootstrap CIs do not overlap ($[-33.4, -29.4]$ vs. $[-28.2, -22.8]$, Table 1). The macrophage-count signal is therefore misleading as well as uninformative: it steers dosing worse than chance. Knowing *how many* macrophages are polarised without knowing *where* the drug, cells, and gradients are is insufficient to solve the indirect, spatially mediated control problem. The choice of observation space can thus make an otherwise learnable task effectively unlearnable. Every spatial-scalar mode, although it does not generalize, still exceeds the random baseline on test (≥ 0 vs. -25.3), so these modes learn *some* transferable structure, far less than the image modes.

3.3 The role of the M1/M2 split and spatial scalars

The full mode family (Appendix D, Table 5) lets us isolate the effect of the explicit M1/M2 polarisation split. Adding it to the cell image (I1→`img_mc_cells_m1m2`) does not improve average OOD test return ($+33.4 \rightarrow +26.0$; the difference is not significant, Welch $p = 0.40$ at $n = 5$), and adding it on top of the full image (I2→I2m) leaves it essentially unchanged ($+32.7 \rightarrow +32.5$). With only five seeds, an equivalence test (TOST, margin $\pm 10 \approx 1\sigma$) does not confirm equivalence for either pair. The data therefore show *no detectable improvement* from the pre-computed split on this toy model, and do not prove a null effect (Appendix E). A plausible mechanism is that, once the substrate fields are spatially available, the CNN can already read polarisation from the `pro_tumoral`/`anti_tumoral` channels. None of the spatial-scalar modes generalize (all within $[-0.0, +9.5]$ test return), whether they carry the M1/M2 split or spatially resolved substrate summaries. Their common failure is the loss of per-cell spatial layout, which the k-means summaries do not preserve.

Appendix F shows a matched example episode comparing an image mode, a spatial-scalar mode, and the POMDP baseline rolled out from an identical network-field initial condition. Without a spatially explicit representation, the agent cannot target the drug precisely enough to convert dose into macrophage repolarisation on an unseen layout. The image mode reaches near-eradication (3 tumor cells), while the scalar and POMDP modes leave 70 and 130 cells respectively.

4 Discussion

The central outcome of this work is PHYSIGYM: a framework that lets RL agents control a full agent-based TME through a standardised interface, without recompilation, and with arbitrary observation and action spaces. The toy-model case study shows the framework end-to-end. An agent learns a spatial drug-delivery policy that moves the microenvironment toward an anti-tumoral state and, in the best episodes, nearly eradicates the tumor (3 of 124 tumor cells left in the matched held-out episode of Appendix F).

The toy model also gives a clear result on observation spaces, which appears only under distribution shift. On the training rectangle layout, spatial-scalar summaries are fully competitive with image observations. On the held-out network-field, image observations generalize to $\approx +33$ test return while the scalar summaries fall to ≈ 0. The CNN in the image modes can compute cross-channel co-localisation statistics (e.g. whether `drug_1` and macrophages overlap at the same location) directly from the pixel layout, and these statistics transfer across geometries. Scalar modes that encode k-means cluster centroids and absolute positions overfit the fixed strip geometry of the training layout. Once cell layout is spatially available, preserving the substrate fields (I2) or pre-computing the M1/M2 split (I2m) does not further improve average transfer over the cell-only image (I1) on this toy model.

Drug aiming under geometry shift. The overfit is visible in the actions of the policies. For every logged episode we compute the mean distance between the drug-injection centre (`action_x`, `action_y`) and the tumour centroid (from the episode initial conditions), averaged over the steps on which a dose is actually applied, and compare this aiming error between the training rectangle and the held-out network-field (Fig. 8). On the training layout all six modes aim equally well (mean normalised error ≈ 0.34–0.36 for both families). Under the geometry shift, image modes hold their aim or slightly improve it (I2: $0.359 \rightarrow 0.343$; I2m: $0.352 \rightarrow 0.333$), whereas the aim of every scalar mode *degrades* (S3s: $0.341 \rightarrow 0.414$; S3m: $0.348 \rightarrow 0.406$) and its step-to-step variance roughly doubles. The scalar policies have learned *where* the rectangle tumour usually lies, and not *how to locate* the tumour. When the network-field moves it, they inject into empty tissue, and the median final tumour count on test rises accordingly (image $\approx$73–99 vs. scalar 107–136). The aggregate return gap thus results from an aiming failure induced by position-absolute features: a centroid that lies between clumps of cells points at empty tissue (Figure 5c).

Reverse direction. To check that the network-field is not simply an easier test layout, we repeated the study with the roles swapped (train on network-field, test on rectangle) and compared image and scalar modes in matched pairs. In the original direction the image mode has the higher held-out return in $11/13$ pairs (mean $\Delta = +27.1$, 95% CI $[+16.3, +36.2]$, sign test $p = 0.01$), and a smaller generalisation gap in $13/13$ ($\Delta = -34.5$, $p = 10^{-4}$). In the reverse direction the return advantage

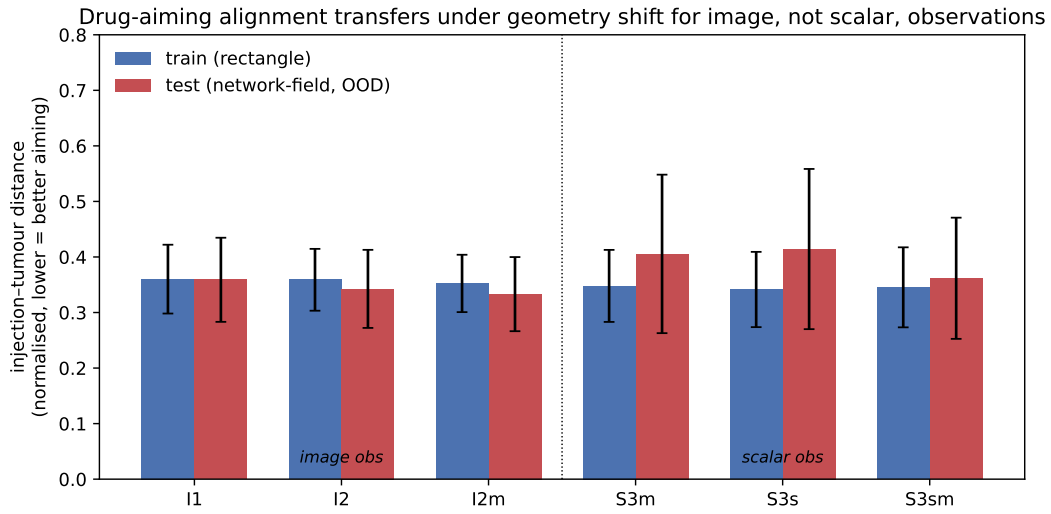


Figure 8: Drug-aiming alignment under geometry shift. Bars show the mean normalised distance between the drug-injection centre and the tumour centroid over dosed steps (lower is better aiming); error bars are the across-episode standard deviation. On the training rectangle (blue) image and scalar observations aim equally well. On the held-out network-field (red) image modes retain their aim while every scalar mode degrades with markedly higher variance. The overfit is a failure to re-target the tumour, and the scalar modes keep their ability to dose.

disappears ($7/15$ pairs, $\Delta = -6.5$, $[-16.9, +4.2]$, $p = 0.70$), and the scalar modes also keep their drug aim under the shift (aiming degradation $+0.008$ against $+0.049$ in the forward direction). Image modes still fit the critic better in both directions (TD residual lower in $13/13$ and $15/15$ pairs). Our interpretation is that scalar agents trained on the irregular network-field cannot memorise a fixed strip position and must use their spatial features, which then transfer. Spatially explicit observations therefore make the agent robust when the training distribution is narrow and structured, a frequent situation in practice (few, stereotyped training geometries). With an irregular training distribution, they give no return advantage in this study.

These observations concern one deliberately simplified model of biological control. In more complex or realistic models, or with different reward designs, action spaces, or RL algorithms, the relative value of cell versus substrate, and of scalar versus image, may differ. PHYSIGYM is designed to make such questions answerable in a reproducible way.

Limitations. The toy model is intentionally small (64 μm, 2D) and cheap to simulate. Scaling to 3D or larger domains will increase the cost of image observations. All experiments use a single RL algorithm (SAC) with one network size per observation family and one reward setting. Alternative algorithms, model-based methods, other reward weightings, or larger encoders may interact differently with observation spaces, and we have not tested whether the image advantage survives these variations. Each mode is trained with a modest, fixed number of seeds ($n = 5$). We report bootstrap confidence intervals on every mean (Table 1, Appendix D), but more seeds would strengthen the conclusions. The image advantage is specific to training on a narrow layout (Section 4). We test only two layouts, and a wider family of train/test distributions would show where between these two extremes it appears. The model's rules and parameters (growth, secretion, chemotaxis, and polarisation rates) are chosen to be biologically plausible and to reproduce the qualitative immunosuppression loop of Section 2.2. They are *illustrative and not fit to a specific tumour or dataset*. We make no quantitative claim about a real cancer, and we have not performed a sensitivity analysis over these parameters. Grounding the rules in measured rates, and verifying that the observation-space conclusion survives this grounding, is important future work. PHYSIGYM is designed to make such studies routine, since the calibration resides in the PHYSICELL configuration and not in the RL code. Complex biological models and 3D models are left to future work.

5 Conclusion

We presented PHYSIGYM, an open-source framework that bridges the PHYSICELL ABM with Gymnasium RL, enabling direct policy learning in spatially explicit biological simulations. Using a TME model with an indirect drug mechanism, we showed that spatially explicit image observations are decisive for *generalization*. With training on a structured rectangle layout and testing on a held-out network-field, image modes retain $\approx+32$ to $+33$ test return, while spatial-scalar summaries, competitive during training, fall to ≈ 0 out of distribution, and a macrophage-count-only POMDP baseline does not learn. The gap concerns generalization and depends on the training distribution: with the layouts swapped, scalar summaries generalize as well as images. Spatially explicit observations therefore protect against overfitting a narrow training distribution. PHYSIGYM provides the infrastructure to study such questions reproducibly, and we expect it to serve as a testbed for the computational biology and RL communities.

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A Generation of Initial Conditions

Rectangle (training layout). Each cell type is placed uniformly in a vertical strip of width 10% of the x-axis, with non-overlapping strip positions sampled from $[0.1, 0.2]$, $[0.3, 0.5]$, $[0.6, 0.8]$ (shuffled across types). This is a highly regular, low-entropy layout.

Network-field (held-out test layout). White Gaussian noise $\eta(\mathbf{x}) \sim \mathcal{N}(0, 1)$ is convolved with a Gaussian kernel G_σ ($\sigma = \ell_c/\sqrt{2}$, $\ell_c = 45\,\mu\text{m}$) to produce a correlated field F. An additional local noise component $\tilde{\xi}$ filtered at width $\ell_c/3$ is added with small weight α to introduce local heterogeneity. The field is normalised to $[0, 1]$ and cells are sampled from positions with $\hat{F} > 0.55$, with probability proportional to $\hat{F}$, producing irregular, clustered configurations that are structurally distinct from the training rectangle.

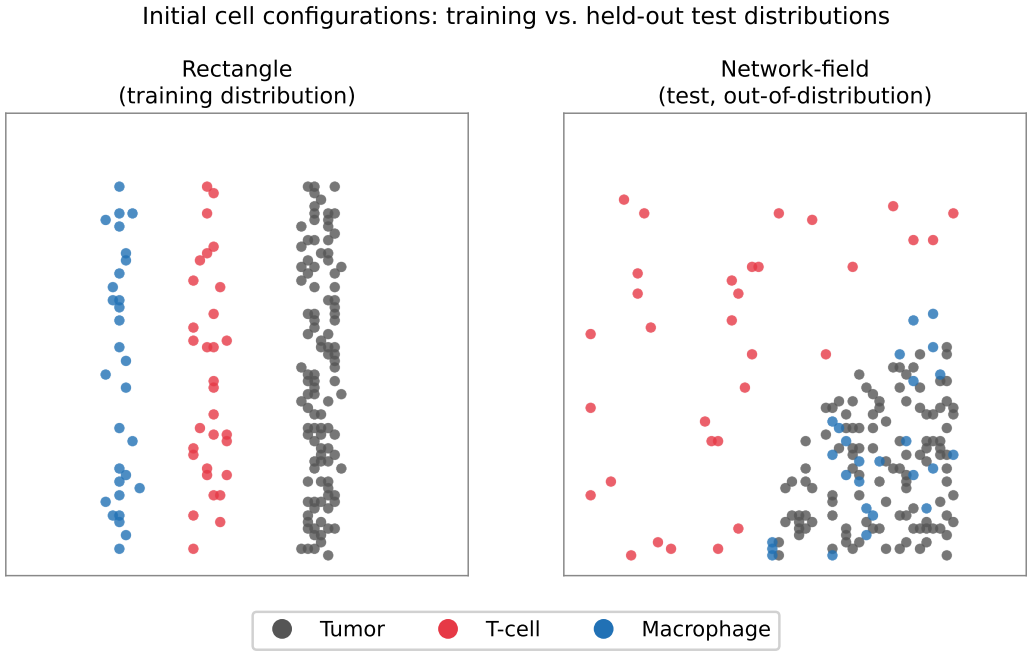


Figure 9: Representative examples of the training (rectangle) and held-out test (network-field) spatial layouts.

B Bridge fidelity

We ran one model configuration and one seed twice: once in standalone PHYSICELL, a process with no Python in it, and once through PHYSIGYM, stepped from Python with the null action so that no drug is released. Both runs are single-threaded, start from the same generated initial condition, and are read from the same on-disk output by the same parser. The two runs do not stop at the same instants: PHYSIGYM ends a step at the first diffusion substep at or past Δt, so with a 0.01-minute substep its second frame sits at 30.01 simulated minutes where the standalone run's sits at 30.00, and

the offset accumulates. We therefore saved the standalone run on the 0.01-minute grid and matched each PHYSIGYM frame to the standalone frame at the identical simulated time (Table 2).

Table 2: PHYSIGYM against standalone PHYSICELL (same seed, same initial condition, one thread, no drug). Cell counts agree for the first 34 of 48 compared frames and first differ at 510 minutes. The position difference starts at one unit in the last place of a double and grows by approximately one order of magnitude every 45 minutes.

t (min)	cells, standalone / bridge	$\max \lvert\Delta\text{position}\rvert$ (μm)
0	183 / 183	0
15	185 / 185	7.1×10^{-15}
60	187 / 187	7.8×10^{-14}
120	188 / 188	2.9×10^{-13}
240	196 / 196	8.2×10^{-10}
360	202 / 202	1.9×10^{-7}
480	206 / 206	4.8×10^{-5}
510	208 / 205	not comparable
control: standalone against standalone, 12,002 frames: bit-identical		

Reproducibility across processes. Two runs launched in separate processes from byte-identical settings, initial condition, and seed, driven by a fixed action schedule for 60 steps (900 simulated minutes), are bitwise identical in cell state and in all five substrate fields when single-threaded (211 vs. 211 cells). With 8 OpenMP threads per run they diverge within the same 60 steps (215 vs. 211 cells, 1 of 5 substrate fields identical).

C Training algorithm and setup

For our control objective (maximal cancer-cell eradication with minimal drug administration) we use Soft Actor-Critic (SAC) [Haarnoja et al., 2018], a state-of-the-art RL algorithm for continuous control. SAC is well suited to our setting due to its high sample efficiency, and as a deep RL method it accommodates different observation types, such as image-based representations, through Convolutional Neural Networks (CNNs) [O'shea and Nash, 2015]. The simulator dominates the computational cost, and we therefore adopt an asynchronous parallel training framework [Mnih et al., 2016, Liu et al., 2025]. A GPU-resident learner updates policy and value networks asynchronously while 13 CPU-resident environment processes generate experience in parallel.

Compute. Each run trains a single observation mode for one seed to 10^5 agent decisions (6×10^5 simulator steps) in roughly 3–5 hours of wall-clock time (including simulation, experience collection, and learner updates) on a single workstation (one NVIDIA GeForce RTX 4090 GPU-resident learner with 13 CPU-resident environment workers on an Intel Core i9-13900K, 32 logical cores). The asynchronous design decouples the GPU learner from the comparatively slow PHYSICELL simulator, so throughput is limited by the CPU environment workers and not by gradient updates. This makes the full observation-space sweep (nine modes, 5 seeds each) tractable on a single machine.

Bridge overhead and throughput. To characterise the cost of the PHYSIGYM bridge, we time a single environment's `step()` in isolation (median over 60 steps after warm-up, one worker, on the workstation above). Each `step()` runs the C++ PHYSICELL simulator step and the Python observation read-back that the bridge adds (Table 3). A single environment sustains $\approx$12–13 steps/s. Switching from the scalar POMDP observation to a full eight-channel image observation raises the per-step time by only $\approx 13\%$ ($76.9 \rightarrow 86.3$ ms), and the cell-only image is indistinguishable from the scalar mode. The observation read-back is a small fraction of the step, which the simulator dominates. The choice of observation space (Section 2.4) therefore has little effect on wall-clock time, and aggregate throughput scales with the number of parallel CPU workers, not with observation richness. The 13-worker configuration reaches on the order of ≈ 150 steps/s of aggregate experience, so the 6×10^5-step budget plus learner updates completes in the 3–5 h per run reported above.

Table 3: Single-environment PHYSIGYM step cost (one CPU worker; median over 60 steps, warm-up discarded). The bridge's observation read-back adds at most $\approx 13\%$ over the raw simulator step; image observations are not a throughput bottleneck.

Observation mode	ms/step (median)	p90 (ms)	steps/s
`scalars_macrophages` (POMDP)	76.9	78.5	13.0
`img_mc_cells` (I1)	75.6	77.6	13.2
`img_mc_cells_substrates` (I2)	86.3	88.3	11.6

Action repeat. Each action selected by the agent is held fixed for 6 consecutive simulator steps (*action repeat* of 6) before the next decision. The controlled process is slow and indirect (drug injection $\rightarrow$ macrophage repolarisation $\rightarrow$ gradient shift $\rightarrow$ T-cell migration $\rightarrow$ tumor death), and a fresh decision at every 15-minute step injects jitter into the drug field without changing the dynamics. The 10^5 agent decisions per run therefore correspond to 6×10^5 simulator steps, the unit of all step axes in Figures 6–7. The value AR$= 6$ comes from a sensitivity sweep (Appendix C.4).

C.1 Asynchronous SAC

Algorithm 1 Asynchronous SAC with Parallel Environments [Liu et al., 2025]

Require: Config d_{arg}, buffer size $|\mathcal{B}|$, LRs, γ, τ
Init shared queues `actor_queue`, `sample_queue`; stop signal `stop_event`
Actor Process (CPU): init $N_{\text{env}} = 13$ envs, local π_{θ_a}
while not `stop_event` **do**
    Receive θ from `actor_queue` if available
    Act: random if $t < T_{\text{start}}$, else π_{θ_a}
    Step envs; push (s, a, r, s', d) to `sample_queue`
end while
Learner Process (GPU): init π_θ, $Q_{\phi_{1,2}}$, targets, $\mathcal{D}$
while total samples $< T$ **do**
    Drain `sample_queue` into $\mathcal{D}$
    for N_{loops} gradient steps **do**
        Sample minibatch; compute SAC target y; update Q, π, α; soft-update targets
    end for
    Push updated θ to `actor_queue`
end while

C.2 Neural Architecture

Scalar modes use a Multi-Layer Perceptron (MLP):

$$\text{FC}(256) \xrightarrow{\text{Mish}} \text{FC}(256)$$

Image modes use an IMPALA-style CNN [Espeholt et al., 2018]. The feature extraction and projection pipeline is defined as:

$$\begin{aligned}
&\text{Conv}(n_c \rightarrow 32,\ k{=}8{\times}8,\ s{=}4) \xrightarrow{\text{Mish}} \\
&\text{Conv}(32 \rightarrow 64,\ k{=}4{\times}4,\ s{=}2) \xrightarrow{\text{Mish}} \\
&\text{Conv}(64 \rightarrow 64,\ k{=}3{\times}3,\ s{=}1) \xrightarrow{\text{Mish}} \text{Flatten} \rightarrow \text{FC}(256) \rightarrow \text{FC}(256)
\end{aligned}$$

C.3 Hyperparameters

Hyperparameter selection. We tuned the environment-side hyperparameters, not the SAC optimiser, and did so once, before and independently of the observation-space comparison. A Latin hypercube design of 20 configurations over the action repeat and the maximum per-step change of each targeted action component (Δx, Δy, Δr) was evaluated with probe rollouts (6 seeds $\times$ 2

Table 4: SAC hyperparameters (TME v2 experiments).

Parameter	Symbol	Value
Agent decisions	T	1×10^5
Action repeat	–	6
Max. per-step action change	$\Delta x, \Delta y, \Delta r$	$0.25,\ 0.25,\ 0.03$
Effective simulator steps	$6T$	6×10^5
Replay buffer size	$\lvert\mathcal{B}\rvert$	2×10^5
Batch size	B	832
Warm-up steps	T_{start}	5,000
Policy / critic LR	η_π, η_Q	3×10^{-4}
Discount factor	γ	0.99
Target smoothing	τ	0.005
Reward weights	$w_{\text{cell}}, w_{\text{dose}}$	$0.3,\ 2.0$
Entropy autotuning	–	enabled
Parallel envs	N_{env}	13
Test frequency	–	every 4 episodes

episodes per configuration) and scored by a standardised composite of tumour reduction, total dose, and action smoothness. This selected action repeat 6 and $(\Delta x, \Delta y, \Delta r) \approx (0.25, 0.25, 0.03)$. The composite does not depend on the reward weights, so the search gives no information about them. The weights $w_{\text{cell}} = 0.3$ and $w_{\text{dose}} = 2$ were set by the argument on their ratio in Section 2.3. The SAC hyperparameters in Table 4 are standard values, shared by every observation mode and not tuned per mode, so no mode benefits from a larger tuning budget than another.

C.4 Choice of action repeat

We select the action-repeat value with a dedicated sensitivity sweep, run independently of the main observation-space experiments. We sweep action_repeat $\in \{1, \ldots, 8\}$ and, for each value, measure three end-of-run quantities aggregated over repeated rollouts (with 95% bootstrap CIs): tumor reduction (control efficacy), an action-smoothness cost $\sum_t \|a_t - a_{t-1}\|^2$ (policy jitter; lower is better), and a composite objective combining tumor reduction, dose, and smoothness (Figure 10).

Two effects drive the choice. (i) *Smoothness*: the action-smoothness cost falls steeply and monotonically with action repeat, from $\approx$110 at AR$=1$ to $\approx$17 at AR$=6$, and then flattens, so most of the stabilisation benefit is already captured by AR$=6$ and larger values add little. This matches the physical timescale of the task: with a 15-minute decision interval on a slow immune process, re-deciding every simulator step injects high-frequency drug-field jitter that does not change the underlying dynamics. (ii) *Composite objective*: the objective is worst at AR$=1$ (-0.88) and rises to a plateau over AR$=5$–8, with AR$=6$ at -0.07. Within that plateau the values lie within each other's confidence intervals, and AR$=6$ is chosen as the smallest repeat that reaches the objective plateau while sitting at the knee of the smoothness curve.

D Full observation-mode comparison

Table 5 reports all nine observation modes, including the `m1m2` variants omitted from the curated main-text Table 1, with $n = 5$ seeds per mode.

Bootstrap confidence intervals. To quantify uncertainty on the *mean* return with $n = 5$ seeds, we resample the seeds of each mode with replacement $B = 10{,}000$ times, recompute the mean end-of-training return of each resample, and report the 2.5th–97.5th percentiles as a 95% confidence interval. The image modes' test CIs lie entirely above $+23$, whereas every spatial-scalar CI *contains zero* (e.g. S3s $[-6.2, +9.1]$) or nearly so. The POMDP baseline's CI $[-33.4, -29.4]$ lies entirely below that of the random policy, $[-28.2, -22.8]$.

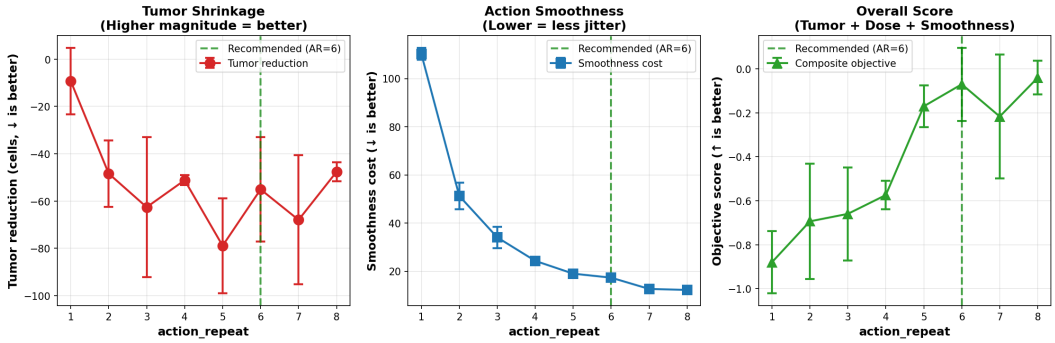


Figure 10: Action-repeat sensitivity sweep (independent of the observation-space experiments; error bars are 95% bootstrap CIs). **Left:** tumor reduction (more negative = more cells removed). **Centre:** action-smoothness cost, which drops sharply and plateaus by AR≈6. **Right:** composite objective (higher is better), which rises to a plateau over AR= 5–8. AR= 6 (dashed) is the smallest repeat on that plateau and at the knee of the smoothness curve.

Table 5: All observation modes ($n = 5$ seeds each; train = rectangle, test = network-field), ordered by test mean. μ: mean end-of-training EWMA-50 return with a 95% bootstrap CI ($B = 10{,}000$ seed resamples).

ID	Mode	Train$_\mu$ (95% CI)	Test$_\mu$ (95% CI)
I1	`img_mc_cells`	$+30.8$ $[+21.3, +38.0]$	$+33.4$ $[+23.6, +43.2]$
I2	`img_mc_cells_substrates`	$+45.3$ $[+40.5, +49.4]$	$+32.7$ $[+28.6, +36.7]$
I2m	`img_mc_cells_substrates_m1m2`	$+40.8$ $[+33.0, +47.3]$	$+32.5$ $[+24.4, +40.5]$
I1m	`img_mc_cells_m1m2`	$+35.0$ $[+22.8, +45.6]$	$+26.0$ $[+14.1, +35.2]$
S5m	`spatial_scalars_cells_spatial_no_scalars_substrates_m1m2`	$+53.6$ $[+25.0, +85.1]$	$+9.5$ $[+0.4, +17.1]$
S3sm	`spatial_scalars_cells_substrates_m1m2`	$+38.3$ $[+20.1, +65.9]$	$+2.6$ $[-6.2, +14.0]$
S3m	`spatial_scalars_cells_m1m2`	$+41.8$ $[+19.8, +68.0]$	$+2.5$ $[-7.6, +11.4]$
S3s	`spatial_scalars_cells_substrates`	$+42.9$ $[+24.7, +58.8]$	-0.0 $[-6.2, +9.1]$
RAND	random policy (baseline)	-18.4 $[-21.9, -14.9]$	-25.3 $[-28.2, -22.8]$
POMDP	`scalars_macrophages`	-20.4 $[-28.6, -13.8]$	-31.8 $[-33.4, -29.4]$

E Significance tests and effect sizes

The bootstrap CIs (Appendix D) quantify uncertainty on each mode's mean; here we add explicit tests so the main claims do not rest on visual CI overlap alone. All tests use the same per-seed end-of-training EWMA-50 test returns ($n = 5$ per mode) and introduce no new runs.

Image vs. spatial-scalar generalization. Pooling the three image modes' per-seed test returns ($n = 15$, mean $+32.8$) against the four spatial-scalar modes' ($n = 20$, mean $+3.7$), a one-sided Mann–Whitney U-test gives $U = 296$, $p = 6.2 \times 10^{-7}$, and Cliff's $\delta = +0.97$, i.e. near-complete rank separation between the families. This is the formal counterpart to the disjoint CIs in Table 5.

M1/M2-split ablation and equivalence test. For the two nested pairs we report both the difference-of-means test and a two-one-sided-tests (TOST) equivalence test against a margin of ± 10 ($\approx 1\sigma$ of a single mode's seed spread). I1 vs. I1m: $+33.4$ vs. $+26.0$, difference not significant (Welch $p = 0.40$) but TOST does *not* establish equivalence ($p = 0.38$). I2 vs. I2m: $+32.7$ vs. $+32.5$, Welch $p = 0.97$, TOST $p = 0.05$ (borderline). At $n = 5$ the data therefore support "no detectable improvement from the pre-computed split" but do not license the stronger claim of statistical equivalence; more seeds would be needed to settle it.

The numbers above are produced by `figures_plotting/stats_tests.py`, which reads the same cached per-seed curves as the tables and figures.

F Example Episode Comparisons

Figure 11 shows a single matched test episode comparing an image mode (`img_mc_cells_substrates`), a spatial-scalar mode (`spatial_scalars_cells_substrates`),

and the POMDP baseline (`scalars_macrophages`) step by step. For an exact comparison, all three agents are rolled out (deterministically, using the policy mean action) from the *identical* held-out network-field initial condition, the same 124-tumor-cell layout replayed from file. The observation mode is therefore the only difference between the three trajectories.

The three regimes separate clearly in this episode. `img_mc_cells_substrates` drives the tumor to near-eradication (3 cells remaining, $+223$ cumulative return). It spends the most drug (4.5 total dose), but places it precisely enough to convert dose into macrophage repolarisation and tumor clearance, with sustained, targeted bursts of dosing (Figure 11, bottom). `spatial_scalars_cells_substrates` achieves partial control (70 cells, $+30$ return, 2.8 dose). It suppresses growth early but then plateaus, because it lacks the spatial substrate feedback needed to target the final phase. `scalars_macrophages` (POMDP) barely acts (0.9 total dose) and the tumor *grows* past its starting size (130 cells, -6.4 return). Seeing only macrophage-polarisation counts, it cannot localise where to inject, so it under-doses and fails, consistent with its aggregate performance at or below the random baseline.

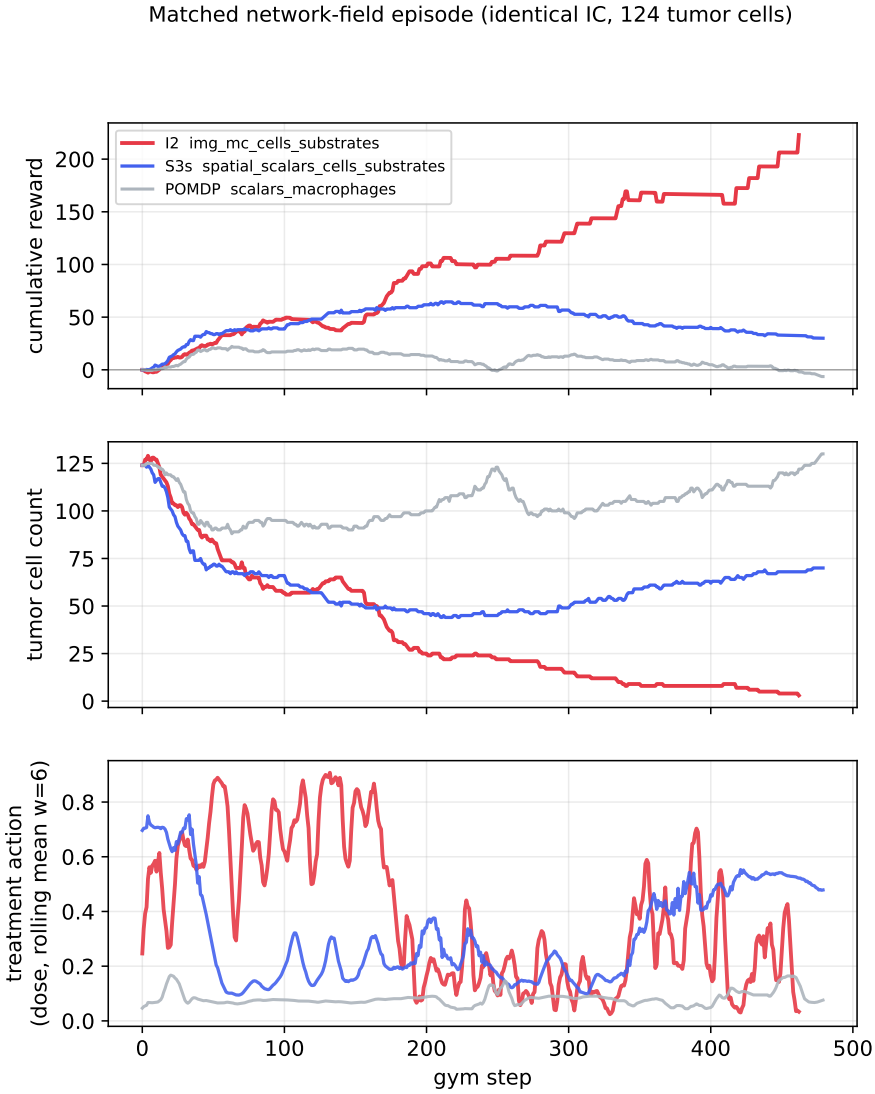


Figure 11: **Matched network-field episode (identical initial condition,** 124 **tumor cells).** Cumulative reward (top), tumor cell count (middle), and the dose action (bottom, rolling mean over 6 steps, which matches the action-repeat horizon used in training, because the raw per-step signal is noisy and unreadable at full resolution) for `img_mc_cells_substrates` (red), `spatial_scalars_cells_substrates` (blue), and the `scalars_macrophages` POMDP baseline (grey), all replayed from the same IC. `img_mc_cells_substrates` reaches near-eradication (3 cells, $+223$ return, 4.5 dose); `spatial_scalars_cells_substrates` achieves partial control (70 cells, $+30$ return, 2.8 dose); the POMDP baseline under-doses and the tumor grows (130 cells, -6.4 return, 0.9 dose). Only the image mode sustains a high, decisive dose action long enough to drive the tumor to near-eradication, whereas `spatial_scalars_cells_substrates` oscillates and eventually lets the tumor regrow. Note that the physical dose delivered to the simulation each step is the concentration action (bottom) multiplied by the voxel volume swept by the independently-controlled injection radius action, so the dose action alone does not linearly predict the physical dose spent.